class09

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Importing Candy Data

We start by getting data from FiveThirtyEights' GitHub repo, and reading it in and making a dataframe for it

```
candydata <- read.csv("candy-data.csv")

candy <-  data.frame(candydata, row.names = 1)
head(candy)
```

```
             chocolate fruity caramel peanutyalmondy nougat crispedricewafer
100 Grand            1      0       1              0      0                1
3 Musketeers         1      0       0              0      1                0
One dime             0      0       0              0      0                0
One quarter          0      0       0              0      0                0
Air Heads            0      1       0              0      0                0
Almond Joy           1      0       0              1      0                0
             hard bar pluribus sugarpercent pricepercent winpercent
100 Grand       0   1        0        0.732        0.860   66.97173
3 Musketeers    0   1        0        0.604        0.511   67.60294
One dime        0   0        0        0.011        0.116   32.26109
One quarter     0   0        0        0.011        0.511   46.11650
Air Heads       0   0        0        0.906        0.511   52.34146
Almond Joy      0   1        0        0.465        0.767   50.34755
```

Q1. How many different candy types are in this dataset?

there are 85 rows in this dataset

```
nrow(candy)
```

```
[1] 85
```

there are 85 different types of candy

Q2. How many fruity candy types are in the dataset?

```
sum(candy$fruity)
```

```
[1] 38
```

there are 38 fruity candies

```
candy["Twix", ]$winpercent
```

```
[1] 81.64291
```

```
library(dplyr)
```

```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
candy |>
  filter(row.names(candy)=="Twix") |>
  select(winpercent)
```

```
     winpercent
Twix   81.64291
```

Q3. What is your favorite candy (other than Twix) in the dataset and what is it's winpercent value?

```
candy["100 Grand", ]$winpercent
```

```
[1] 66.97173
```

100 Grand and it has a winpercent of 66.97173

Q4. What is the winpercent value for "Kit Kat"?

```
candy["Kit Kat", ]$winpercent
```

```
[1] 76.7686
```

Kit Kat has a winpercent value of 76.7686

Q5. What is the winpercent value for "Tootsie Roll Snack Bars"?

Toostie Roll has a winpercent value of 49.653503

```
library("skimr")
skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyalmondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedricewafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

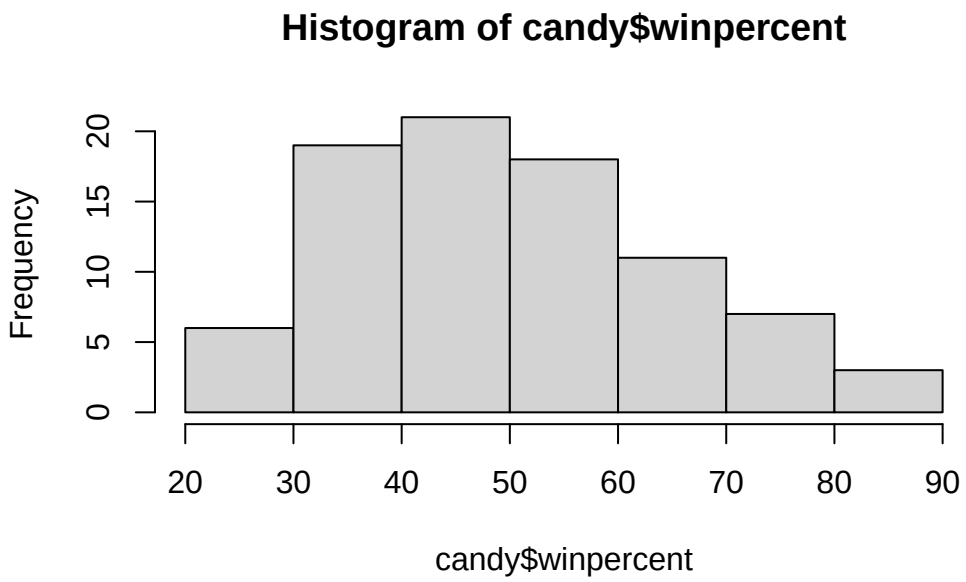
Yes, winpercent operates on a much larger scale than the rest of the data in the skim function. The rest operate on a 0-1 scale whereas winpercent looks like it goes 0-100.

Q7. What do you think a zero and one represent for the candy$chocolate column?

0 represents not containing chocalate whereas 1 represents containing chocolate candy.

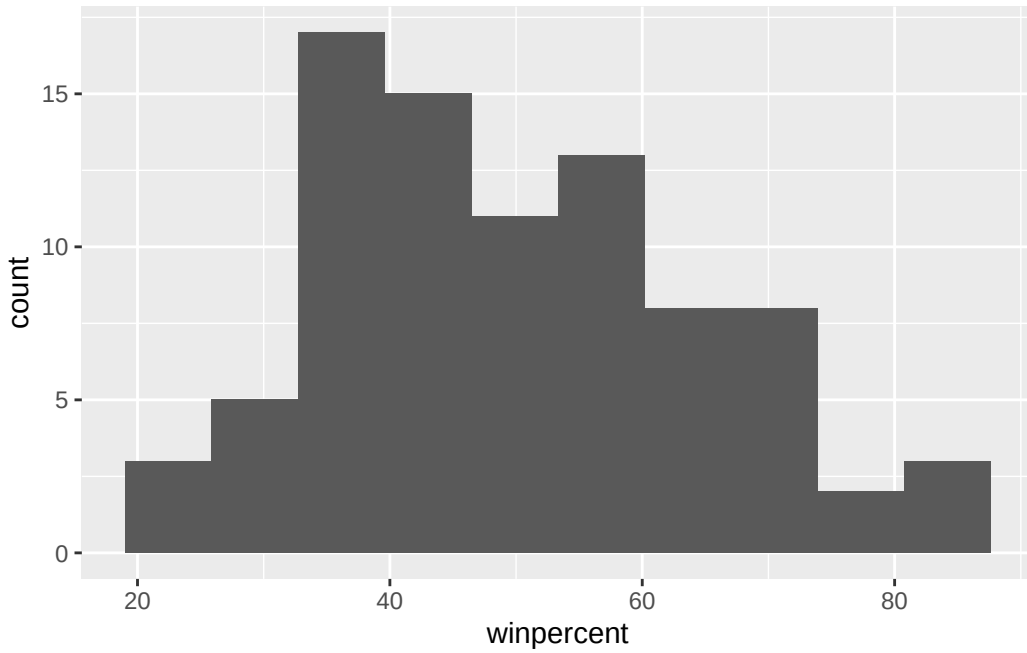
Q8. Plot a histogram of winpercent values using both base R an ggplot2.

```
rcandyhist <- hist(candy$winpercent)
```




```
library(ggplot2)

ggcandyhist <- ggplot(candy, aes(x = winpercent)) +
  geom_histogram(bins=10)
ggcandyhist
```



Q9. Is the distribution of winpercent values symmetrical?

Not quite, there is a slight left skew.

Q10. Is the center of the distribution above or below 50%?

```
summary(candy$winpercent)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
  22.45   39.14   47.83   50.32   59.86   84.18 
```

below 50%, median is at 47.83

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

Steps to solve this: 1. Find all choclate candy in the dataset 2. extract or find their winpercent values 3. calculate the mean of these values

4. find all fruit candy
5. find their winpercent values
6. calculate their mean value

```r
head(candy)
```

```
             chocolate fruity caramel peanutyalmondy nougat crispedricewafer
100 Grand            1      0       1              0      0                1
3 Musketeers         1      0       0              0      1                0
One dime             0      0       0              0      0                0
One quarter          0      0       0              0      0                0
Air Heads            0      1       0              0      0                0
Almond Joy           1      0       0              1      0                0
             hard bar pluribus sugarpercent pricepercent winpercent
100 Grand       0   1        0        0.732        0.860   66.97173
3 Musketeers    0   1        0        0.604        0.511   67.60294
One dime        0   0        0        0.011        0.116   32.26109
One quarter     0   0        0        0.011        0.511   46.11650
Air Heads       0   0        0        0.906        0.511   52.34146
Almond Joy      0   1        0        0.465        0.767   50.34755
```

```r
choco <- candy[candy$chocolate==1,]
choco.win <- choco$winpercent
fruit <- candy[candy$fruity==1,]
fruit.win <- fruit$winpercent
mean(choco.win)
```

```
[1] 60.92153
```

```r
mean(fruit.win)
```

```
[1] 44.11974
```

```r
t.test(choco.win, fruit.win)
```

```
    Welch Two Sample t-test

data:  choco.win and fruit.win
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
```

```
sample estimates:
mean of x mean of y
 60.92153  44.11974
```

Chocolate candy is ranked higher than fruit candy on average.

Q12. Is this difference statistically significant?

Yes, p = smaller than .05 so null can be rejected. meaning it is statistically significant.

Q13. What are the five least liked candy types in this set?

```r
inds <- order(candy$winpercent)
head(candy[inds,], 5)
```

```
                   chocolate fruity caramel peanutyalmondy nougat
Nik L Nip                  0      1       0              0      0
Boston Baked Beans         0      0       0              1      0
Chiclets                   0      1       0              0      0
Super Bubble               0      1       0              0      0
Jawbusters                 0      1       0              0      0
                   crispedricewafer hard bar pluribus sugarpercent pricepercent
Nik L Nip                         0    0   0        1        0.197        0.976
Boston Baked Beans                0    0   0        1        0.313        0.511
Chiclets                          0    0   0        1        0.046        0.325
Super Bubble                      0    0   0        0        0.162        0.116
Jawbusters                        0    1   0        1        0.093        0.511
                   winpercent
Nik L Nip            22.44534
Boston Baked Beans   23.41782
Chiclets             24.52499
Super Bubble         27.30386
Jawbusters           28.12744
```

Nik-L-Nip, Boston Baked Beans, Chiclets, Super Buble, Jawbusters.

Q14. What are the top 5 all time favorite candy types out of this set?

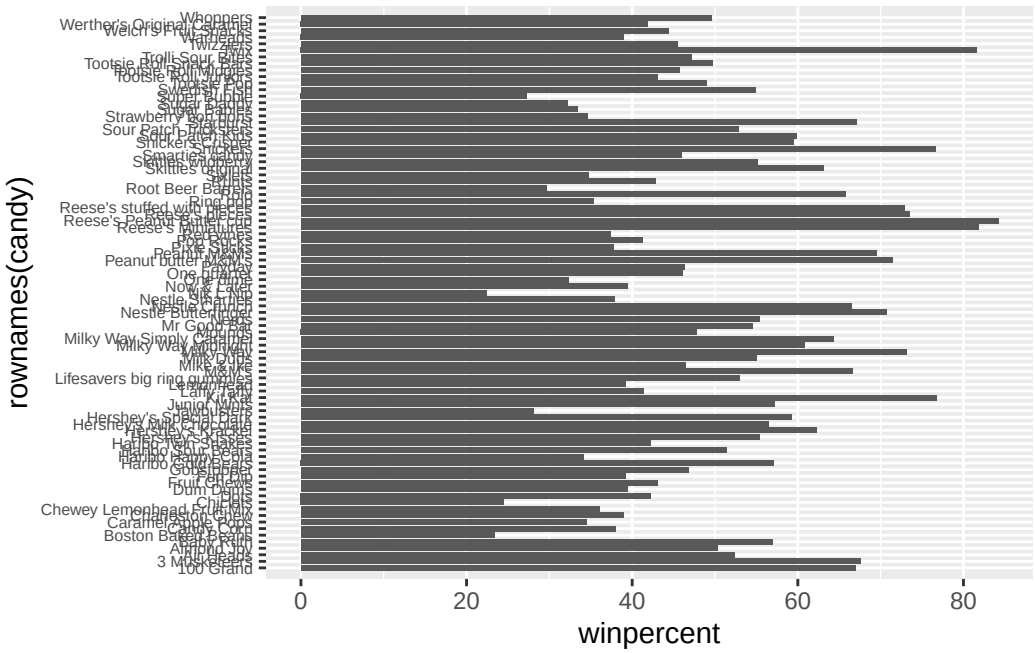
```r
tail(candy[inds,], 5)
```

```
                          chocolate fruity caramel peanutyalmondy nougat
Snickers                          1      0       1              1      1
Kit Kat                           1      0       0              0      0
Twix                              1      0       1              0      0
Reese's Miniatures                1      0       0              1      0
Reese's Peanut Butter cup         1      0       0              1      0
                          crispedricewafer hard bar pluribus sugarpercent
Snickers                                  0    0   1        0        0.546
Kit Kat                                   1    0   1        0        0.313
Twix                                      1    0   1        0        0.546
Reese's Miniatures                        0    0   0        0        0.034
Reese's Peanut Butter cup                 0    0   0        0        0.720
                          pricepercent winpercent
Snickers                         0.651   76.67378
Kit Kat                          0.511   76.76860
Twix                             0.906   81.64291
Reese's Miniatures               0.279   81.86626
Reese's Peanut Butter cup        0.651   84.18029
```

Snickers, Kit Kat, Twiz, Reese's Miniatures, Resse's Peanut Butter cup

Q15. Make a first barplot of candy ranking based on winpercent values.

```r
ggplot(candy) +
  aes(winpercent, rownames(candy)) +
  geom_col() +
  theme(axis.text.y = element_text(size = 6)) +
  scale_y_discrete(expand = expansion(mult = .02))
```




```
ggsave("barplot1.png", height = 10, width = 6)
```

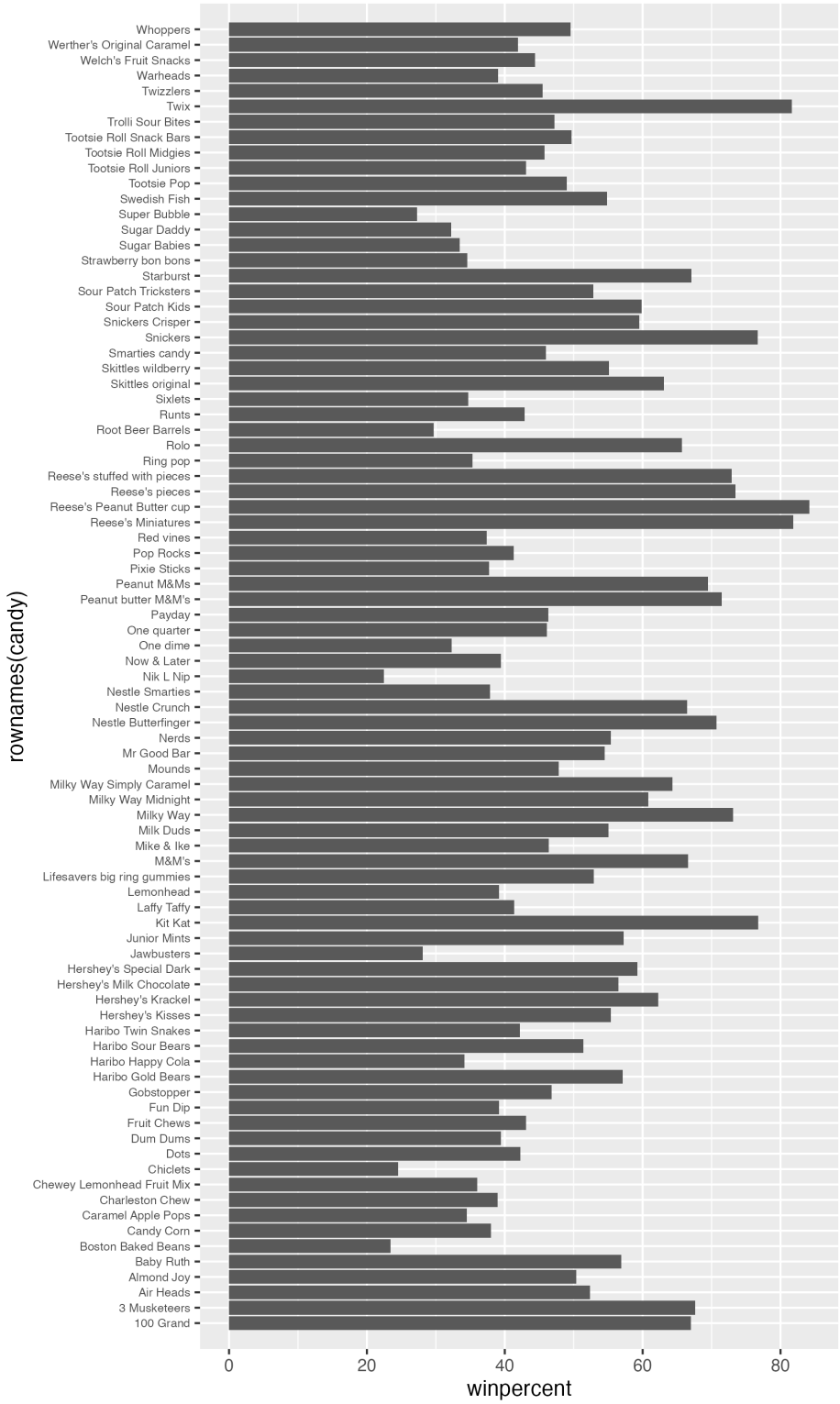
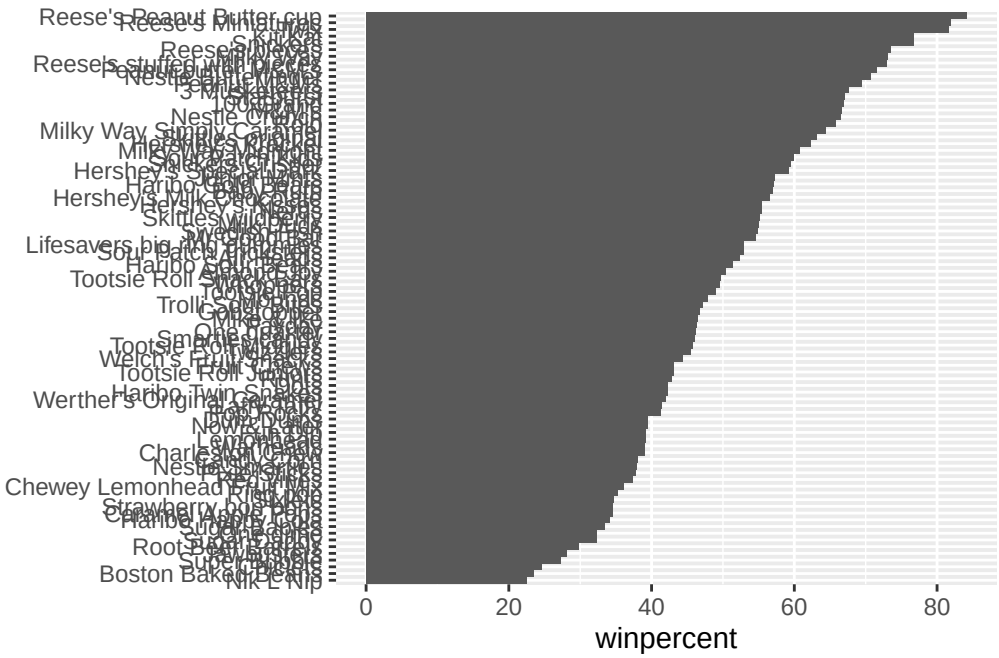


Figure 1: My first Barplot for Q15

Q16. This is quite ugly, use the reorder() function to get the bars sorted by winpercent?

```
ggplot(candy) +
  aes(winpercent,
      reorder(rownames(candy), winpercent)) +
  geom_col() +
    ylab("")
```




```
ggsave("barplot2.png", height = 10, width = 6)
```

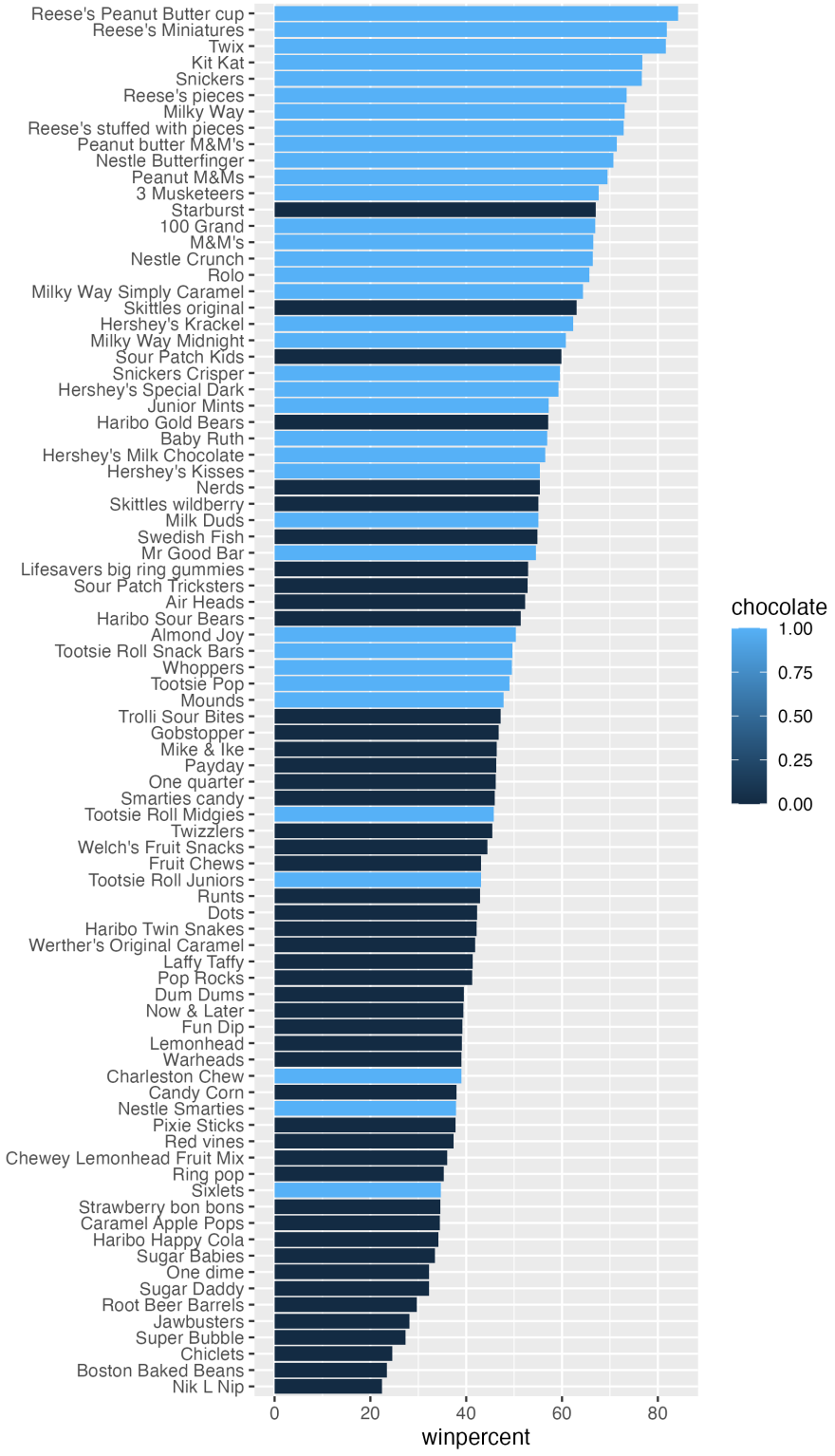
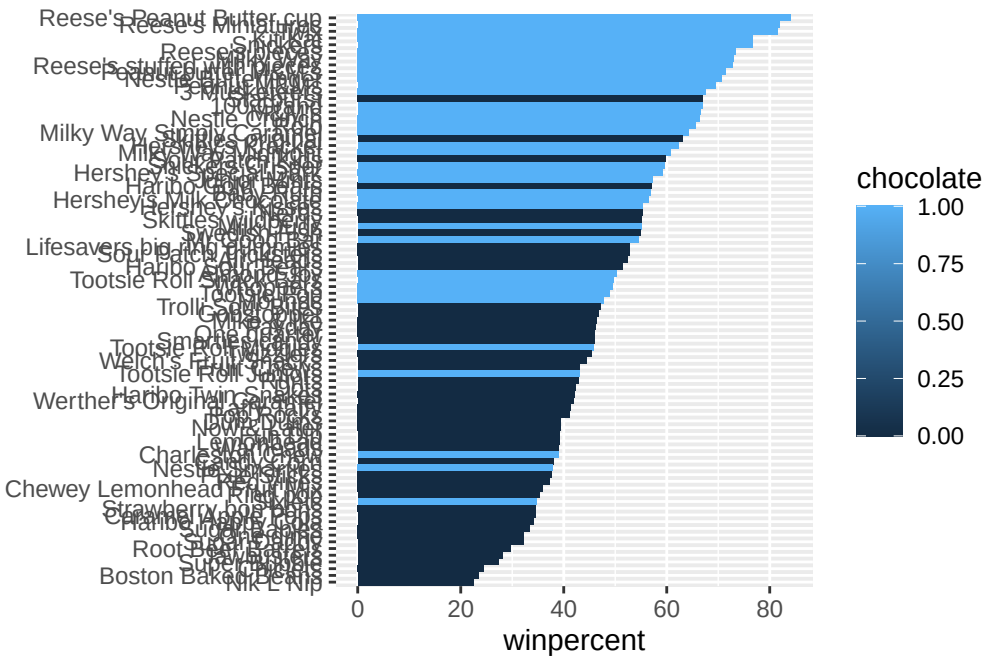


Figure 2: My second Barplot for Q16

##Times to add some useful color

```
ggplot(candy) +
  aes(winpercent,
      reorder(rownames(candy), winpercent), fill = chocolate) +
  geom_col() +
    ylab("")
```




```
ggsave("barplot2.png", height = 10, width = 6)
```

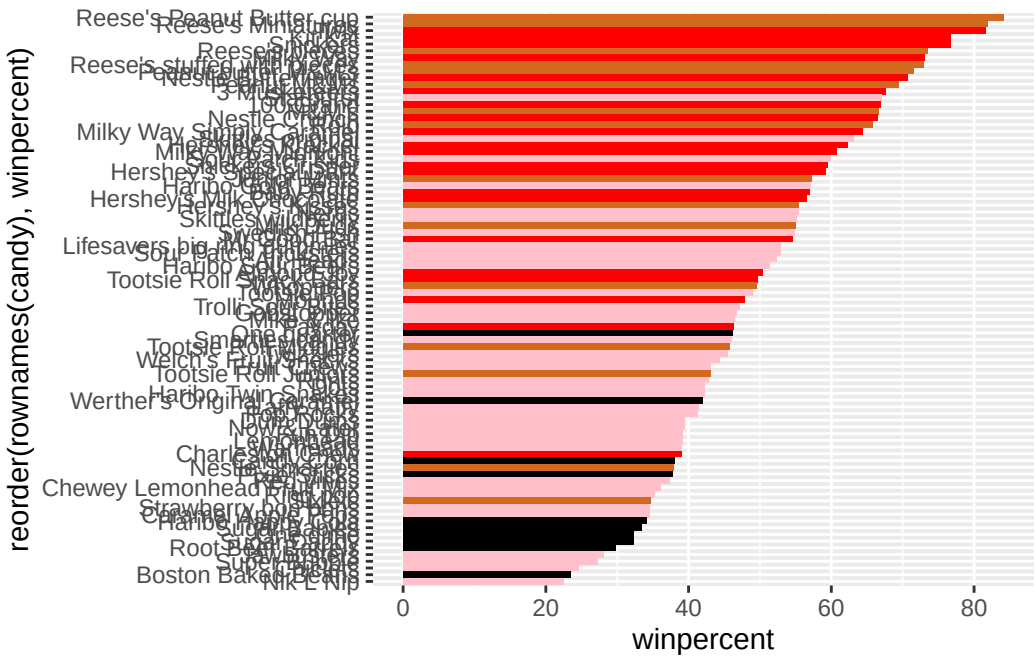
I want custom colors that I pick so we need to makes this ourselves…

```
my_cols <- rep("black", nrow(candy))
my_cols[candy$chocolate ==1 ] <- "chocolate"
my_cols[candy$bar == 1] <- "red"
my_cols[candy$fruity == 1] <- "pink"
my_cols
```

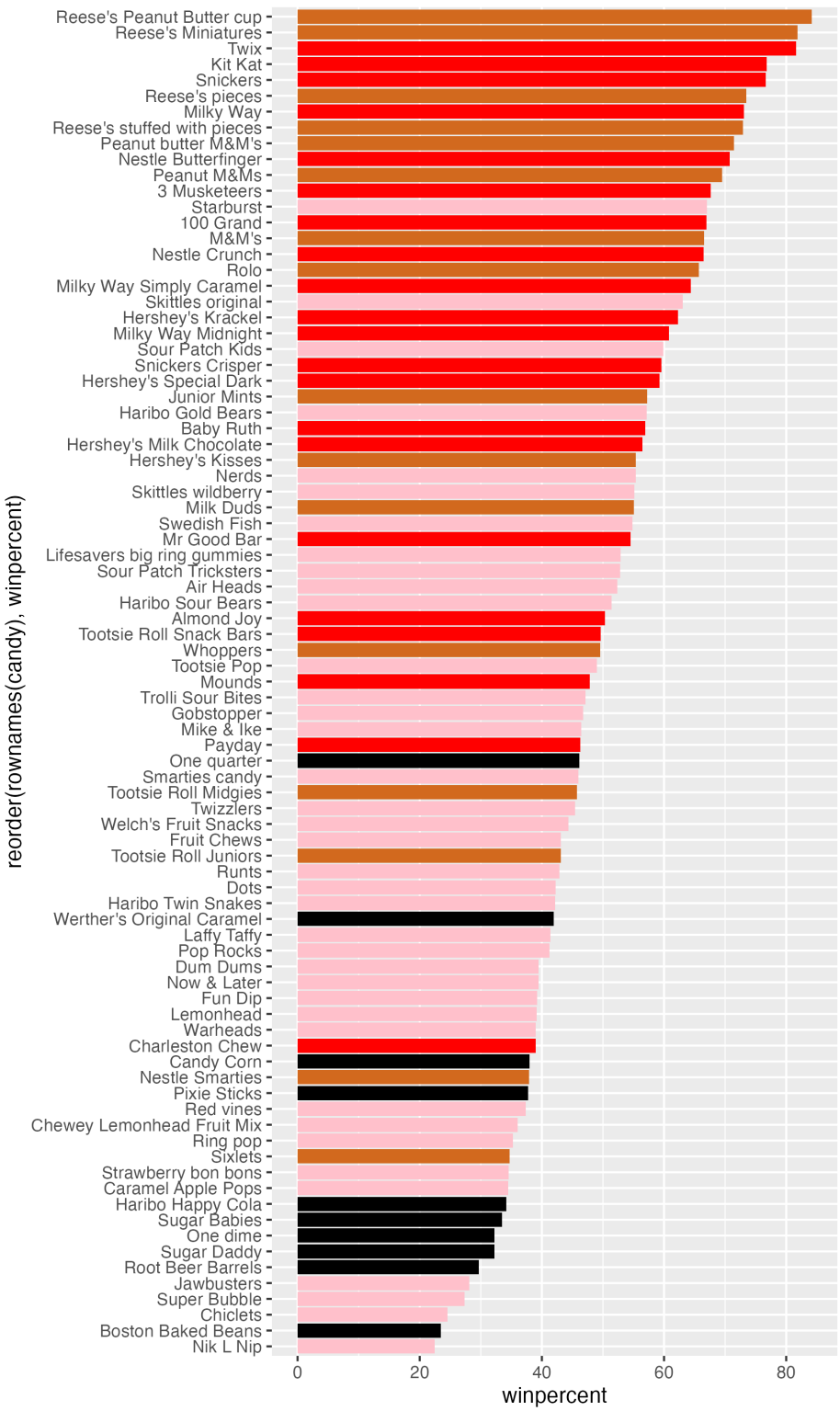
```
 [1] "red"       "red"       "black"     "black"     "pink"      "red"
 [7] "red"       "black"     "black"     "pink"      "red"       "pink"
[13] "pink"      "pink"      "pink"      "pink"      "pink"      "pink"
```

```
[19] "pink"      "black"     "pink"      "pink"      "chocolate" "red"
[25] "red"       "red"       "pink"      "chocolate" "red"       "pink"
[31] "pink"      "pink"      "chocolate" "chocolate" "pink"      "chocolate"
[37] "red"       "red"       "red"       "red"       "red"       "pink"
[43] "red"       "red"       "pink"      "pink"      "red"       "chocolate"
[49] "black"     "pink"      "pink"      "chocolate" "chocolate" "chocolate"
[55] "chocolate" "pink"      "chocolate" "black"     "pink"      "chocolate"
[61] "pink"      "pink"      "chocolate" "pink"      "red"       "red"
[67] "pink"      "pink"      "pink"      "pink"      "black"     "black"
[73] "pink"      "pink"      "pink"      "chocolate" "chocolate" "red"
[79] "pink"      "red"       "pink"      "pink"      "pink"      "black"
[85] "chocolate"
```

```
ggplot(candy) +
  aes(winpercent,
      reorder(rownames(candy), winpercent), fill = chocolate) +
  geom_col(fill = my_cols)
```




```
ggsave("barplot3.png", width = 6, height = 10)
```



>Q17. What is the

worst ranked chocolate candy?

Sixlets

> Q18. What is the best ranked fruity candy?

Starbursts

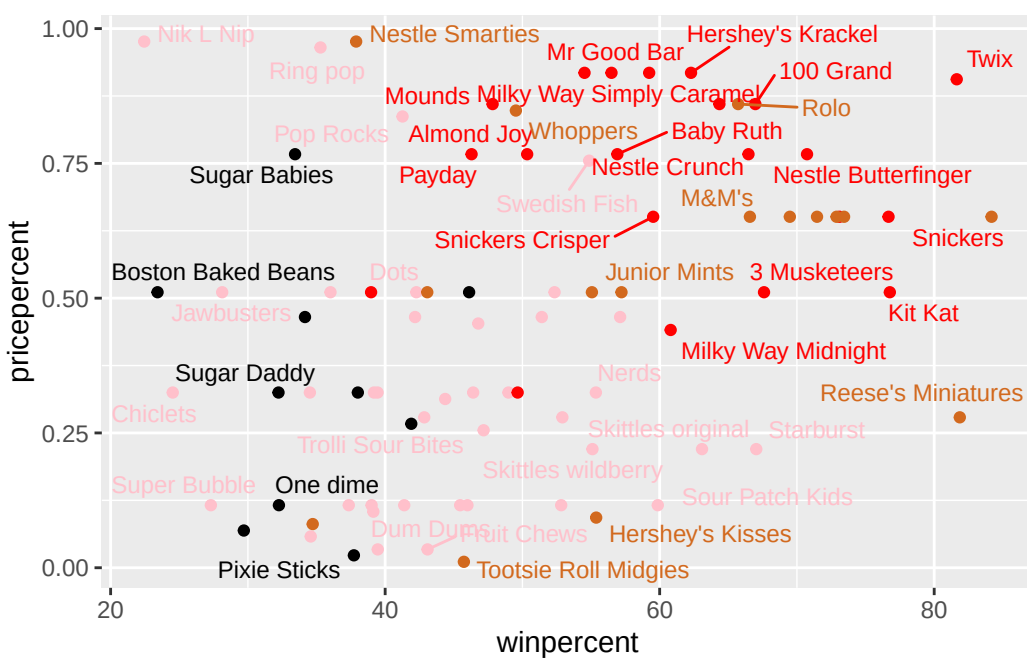
Taking a look at pricepercent

We can use the **ggrepel** package for better label placement:

```
library(ggrepel)

# How about a plot of win vs price
ggplot(candy) +
  aes(winpercent, pricepercent, label=rownames(candy)) +
  geom_point(col=my_cols) +
  geom_text_repel(col=my_cols, size=3, max.overlaps = 10)
```

```
Warning: ggrepel: 40 unlabeled data points (too many overlaps). Consider
increasing max.overlaps
```



> Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

Reese's Miniature

> Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

```
ord <- order(candy$pricepercent, decreasing = TRUE)
head( candy[ord,c(11,12)], n=5 )
```

```
                         pricepercent winpercent
Nik L Nip                       0.976   22.44534
Nestle Smarties                 0.976   37.88719
Ring pop                        0.965   35.29076
Hershey's Krackel               0.918   62.28448
Hershey's Milk Chocolate        0.918   56.49050
```

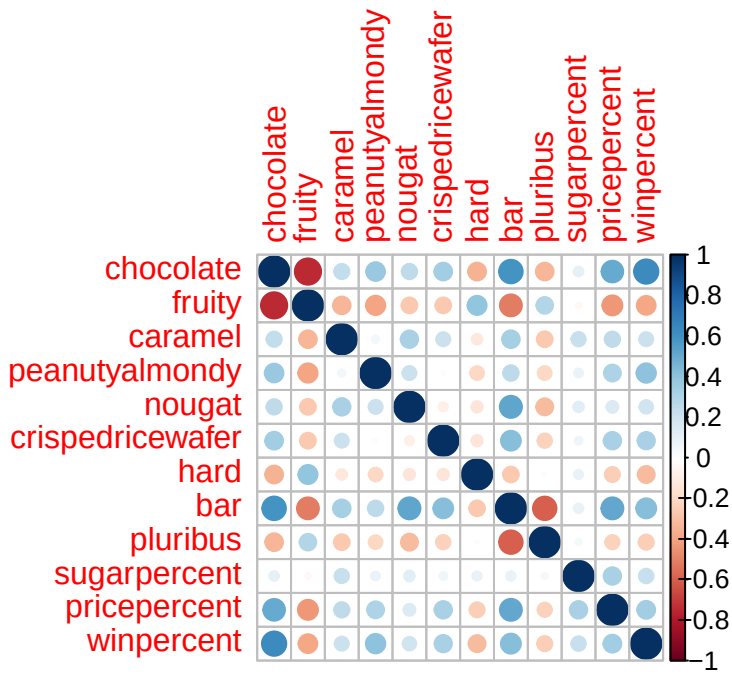
Nik L Nip, Nestle Smarties, Ring pop, Hershey's Krackel, Hershey's Milk Chocolate. Nik L Nip's are the least popular. ## Exploring the correlation structure

Pearson correlation values range from -1 to +1

```
library(corrplot)
```

```
corrplot 0.95 loaded
```

```
cij <- cor(candy)
corrplot(cij)
```




```r
cor(candy)
```

```
                  chocolate      fruity     caramel peanutyalmondy      nougat
chocolate         1.0000000 -0.74172106  0.24987535     0.37782357  0.25489183
fruity           -0.7417211  1.00000000 -0.33548538    -0.39928014 -0.26936712
caramel           0.2498753 -0.33548538  1.00000000     0.05935614  0.32849280
peanutyalmondy    0.3778236 -0.39928014  0.05935614     1.00000000  0.21311310
nougat            0.2548918 -0.26936712  0.32849280     0.21311310  1.00000000
crispedricewafer  0.3412098 -0.26936712  0.21311310    -0.01764631 -0.08974359
hard             -0.3441769  0.39067750 -0.12235513    -0.20555661 -0.13867505
bar               0.5974211 -0.51506558  0.33396002     0.26041960  0.52297636
pluribus         -0.3396752  0.29972522 -0.26958501    -0.20610932 -0.31033884
sugarpercent      0.1041691 -0.03439296  0.22193335     0.08788927  0.12308135
pricepercent      0.5046754 -0.43096853  0.25432709     0.30915323  0.15319643
winpercent        0.6365167 -0.38093814  0.21341630     0.40619220  0.19937530
                 crispedricewafer        hard          bar    pluribus
chocolate              0.34120978 -0.34417691  0.59742114 -0.33967519
fruity                -0.26936712  0.39067750 -0.51506558  0.29972522
caramel                0.21311310 -0.12235513  0.33396002 -0.26958501
peanutyalmondy        -0.01764631 -0.20555661  0.26041960 -0.20610932
nougat                -0.08974359 -0.13867505  0.52297636 -0.31033884
crispedricewafer       1.00000000 -0.13867505  0.42375093 -0.22469338
```

```
hard                  -0.13867505  1.00000000 -0.26516504  0.01453172
bar                    0.42375093 -0.26516504  1.00000000 -0.59340892
pluribus              -0.22469338  0.01453172 -0.59340892  1.00000000
sugarpercent           0.06994969  0.09180975  0.09998516  0.04552282
pricepercent           0.32826539 -0.24436534  0.51840654 -0.22079363
winpercent             0.32467965 -0.31038158  0.42992933 -0.24744787
                 sugarpercent pricepercent winpercent
chocolate          0.10416906    0.5046754  0.6365167
fruity            -0.03439296   -0.4309685 -0.3809381
caramel            0.22193335    0.2543271  0.2134163
peanutyalmondy     0.08788927    0.3091532  0.4061922
nougat             0.12308135    0.1531964  0.1993753
crispedricewafer   0.06994969    0.3282654  0.3246797
hard               0.09180975   -0.2443653 -0.3103816
bar                0.09998516    0.5184065  0.4299293
pluribus           0.04552282   -0.2207936 -0.2474479
sugarpercent       1.00000000    0.3297064  0.2291507
pricepercent       0.32970639    1.0000000  0.3453254
winpercent         0.22915066    0.3453254  1.0000000
```

> Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

Fruity and chocolate candy.

> Q23. Similarly, what two variables are most positively correlated?

Chocolate and Bar

##Principal Component Analysis

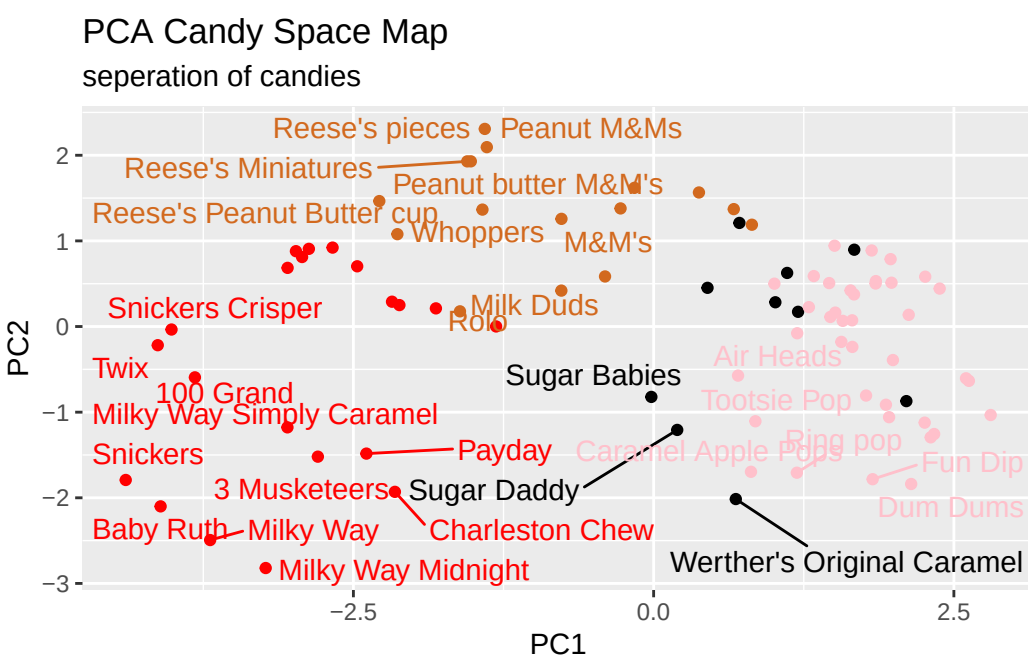
```r
pca <- prcomp(candy, scale = T)
summary(pca)
```

```
Importance of components:
                          PC1    PC2    PC3     PC4    PC5     PC6     PC7
Standard deviation     2.0788 1.1378 1.1092 1.07533 0.9518 0.81923 0.81530
Proportion of Variance 0.3601 0.1079 0.1025 0.09636 0.0755 0.05593 0.05539
Cumulative Proportion  0.3601 0.4680 0.5705 0.66688 0.7424 0.79830 0.85369
                           PC8     PC9    PC10    PC11    PC12
Standard deviation     0.74530 0.67824 0.62349 0.43974 0.39760
Proportion of Variance 0.04629 0.03833 0.03239 0.01611 0.01317
Cumulative Proportion  0.89998 0.93832 0.97071 0.98683 1.00000
```

The main results figure: the PCA score plot:

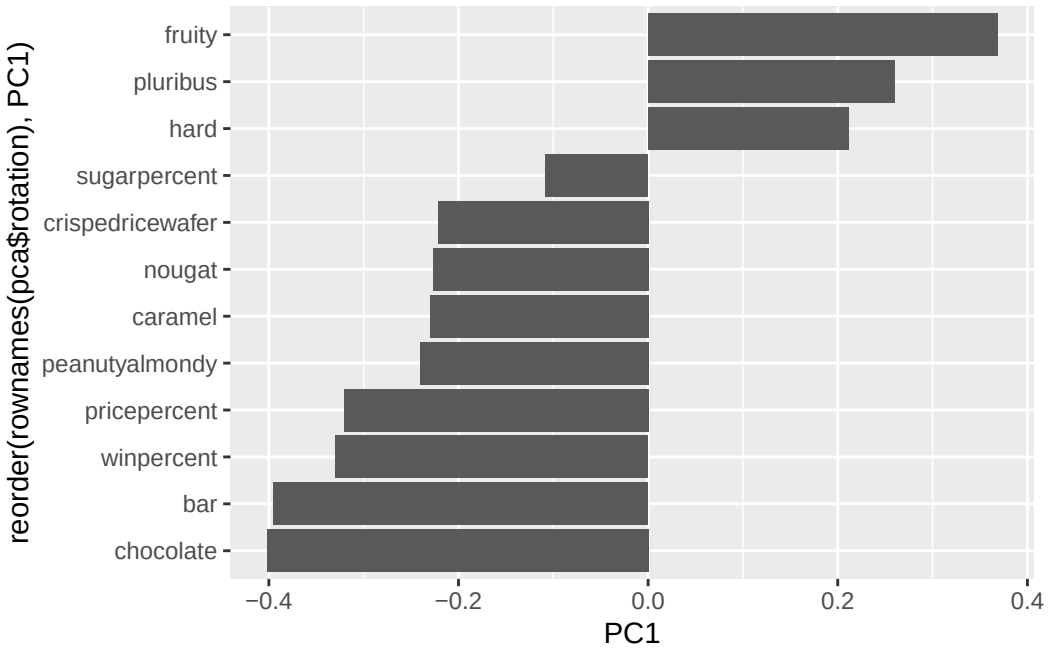
```
ggplot(pca$x) +
  aes(PC1,PC2, label=rownames(pca$x)) +
  geom_point(col = my_cols) +
  geom_text_repel(col=my_cols) +
  labs(title="PCA Candy Space Map", subtitle = "seperation of candies")
```

```
Warning: ggrepel: 56 unlabeled data points (too many overlaps). Consider
increasing max.overlaps
```



The "loadings" plot for PC1

```
ggplot(pca$rotation) +
  aes(x = PC1, y = reorder(rownames(pca$rotation), PC1)) +
  geom_col()
```




```
#library(plotly)
#ggplotly(p)
```

> Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

Fruity, Pluribus and hard are all picked up strongly positively in PC1. Yes this makes sense, we saw this in the PCA score plot as well.

> Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a "winning" candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

Chocolate & Caramel, Fruity & Pluribas, Fruity & Hard. PCA visualization puts the candies into visualizable groups, showing what candies share characteristics and which ones differ. The correlation analyses gives numerical values to how strong two factors are associated with each other and whether they contribute positively or negatively to each others likeability.